

PROJECT REPORT

APPLE FRUIT SEGMENTATION

Submitted by

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1.TITLE

Apple Fruit Segmentation Using Color-Based Thresholding and Morphological Image Processing

2. PROBLEM STATEMENT

Identifying and separating an apple from its background is an important task in digital image processing. It can be used in fruit quality inspection, agricultural automation, robotic harvesting, and computer vision applications.

In this project, apple images are segmented from their backgrounds using two image-processing techniques:

1. Otsu's thresholding
2. HSV color-based thresholding followed by morphological processing

The performance of these techniques is compared using quantitative evaluation metrics and visual inspection.

3. OBJECTIVES

The major objectives of this project are:

- To understand the problem of apple image segmentation.
- To collect and prepare apple images and their corresponding ground-truth masks.
- To apply Otsu's thresholding for image segmentation.
- To apply HSV color-based thresholding for detecting apple regions.
- To remove noise and fill small holes using morphological operations.
- To generate segmented apple images.
- To compare the performance of Otsu thresholding and HSV-based segmentation.

- To evaluate the segmentation results using IoU, Dice Score, and Pixel Accuracy.
- To test the selected method on wild apple images.
- To identify the limitations and possible improvements of the system.

4. DATASET / IMAGE DESCRIPTION

Dataset Description

The project uses an apple image dataset containing apple images and corresponding segmentation masks.

The dataset folders contain:

- Original/object images: 64 image files
- Mask images: 71 mask files
- Wild apple images: A separate set of apple images without ground-truth masks

The original images contain apples captured against different backgrounds. The mask images represent the expected apple regions and are used as ground-truth data for evaluating the segmentation methods.

The original images and masks were paired using their common filename identifiers. The images were then processed using Python and OpenCV.

Image Characteristics

- Images are stored mainly in JPG and PNG formats.
- The images are colour images.
- The apple regions have different colours, including red and orange shades.
- Some images contain background regions with colours similar to the apple.
- The wild images contain more natural variations in lighting, background, and apple appearance.

Preprocessing

The following preprocessing steps were performed:

- The dataset folders were inspected.
- Original images and mask files were identified.
- Images were paired with their corresponding masks.
- Images were read using OpenCV.
- Images were converted from BGR format to RGB format for displaying results.
- Images were converted from BGR to grayscale for Otsu thresholding.
- Images were converted from BGR to HSV colour space for colour-based segmentation.

Dataset Limitations

- The ground-truth masks are available only for the selected dataset images.
- The wild apple images do not have ground-truth masks.
- The dataset contains variations in lighting and background.
- Some background regions may have colours similar to the apple.
- The selected HSV thresholds may not work equally well for all apple colours.

5. METHODOLOGY

The overall workflow of the project is:

Input Image → Preprocessing → Otsu Segmentation / HSV Segmentation → Morphological Processing → Segmented Image → Comparison → Evaluation

Stage 1: Input Image

An apple image is selected from the dataset. The image contains an apple and its surrounding background.

Stage 2: Preprocessing

The input image is read using OpenCV. Different colour representations are created according to the selected segmentation method.

- A grayscale image is generated for Otsu thresholding.
- An HSV image is generated for colour-based thresholding.

HSV separates hue, saturation, and brightness information. This makes it useful for identifying apple regions based on their colour.

Stage 3: Otsu Thresholding

Otsu's thresholding is applied to the grayscale image.

Otsu's algorithm automatically selects a threshold value by separating the image into two classes:

- Foreground
- Background

The resulting binary image is used as the Otsu segmentation output.

Stage 4: HSV Colour-Based Segmentation

The image is converted into HSV colour space. HSV ranges are selected to identify red and orange apple regions.

Two colour ranges are used because red may appear near both ends of the hue scale:

- A lower red/orange hue range
- An upper red hue range

The two binary masks are combined using a bitwise OR operation. This produces a mask containing the detected apple-coloured regions.

Stage 5: Morphological Processing

Morphological operations are applied to improve the HSV segmentation mask.

A (5×5) kernel is used.

Opening

Opening is performed using erosion followed by dilation. It helps to:

- Remove small noise.
- Remove isolated unwanted pixels.
- Smooth small irregularities.

Closing

Closing is performed using dilation followed by erosion. It helps to:

- Fill small holes.
- Connect nearby regions.
- Improve the continuity of the apple mask.

Stage 6: Segmented Image Generation

The final binary mask is applied to the original image using a bitwise AND operation. This keeps the detected apple region and removes or darkens the background.

The result is a segmented apple image.

Stage 7: Comparison

The outputs of the two methods are compared:

- Original image
- Ground-truth mask
- Otsu segmentation
- HSV segmentation with morphological processing

A comparison graph is also created to display the evaluation scores.

Stage 8: Evaluation

The segmentation results are evaluated using the following metrics:

Intersection over Union

IoU, also called Jaccard Index, measures the overlap between the predicted mask and the ground-truth mask.

A higher IoU indicates better overlap.

Dice Score

Dice Score measures the similarity between the predicted mask and the ground-truth mask.

A higher Dice Score indicates better segmentation similarity.

Pixel Accuracy

Pixel Accuracy measures the percentage of correctly classified pixels.

Pixel Accuracy is useful, but it may be affected by the large number of background pixels. Therefore, IoU and Dice Score are also considered.

6. IMPLEMENTATION

Programming Language

The project was implemented using Python.

Libraries Used

The following libraries were used:

- **NumPy:** Numerical operations and array processing.

- **OpenCV:** Image reading, colour conversion, thresholding, masking, and morphological operations.
- **Matplotlib:** Displaying original images, masks, segmented images, and comparison graphs.
- **Pandas:** Storing and exporting the evaluation results into a CSV file.
- **Google Colab:** Development and execution environment.
- **Google Drive:** Dataset and result storage.

Implementation Details

The important implementation steps include:

- Reading original images and masks.
- Matching image files with their corresponding mask files.
- Converting images into grayscale and HSV colour spaces.
- Applying Otsu's thresholding.
- Applying HSV colour thresholding.
- Combining colour masks.
- Applying morphological opening and closing.
- Generating segmented images.
- Calculating IoU, Dice Score, and Pixel Accuracy.
- Saving the comparison table as a CSV file.
- Creating and saving a comparison graph.
- Testing the HSV-based method on wild apple images.

Techniques Used

☐ Image Preprocessing

- Image reading using OpenCV.
- Conversion between BGR, RGB, grayscale, and HSV colour spaces.

□ **Thresholding**

- Otsu's automatic thresholding.
- HSV colour-based thresholding.

□ **Binary Image Segmentation**

- Creation of binary masks to separate the apple region from the background.

□ **Morphological Image Processing**

- Opening for removing small noise.
- Closing for filling small holes and connecting regions.
- A 5×5 structuring element was used.

□ **Masking**

- Bitwise AND operation was used to extract the segmented apple from the original image.

□ **Segmentation Evaluation**

- Intersection over Union (IoU/Jaccard Index)
- Dice Score
- Pixel Accuracy

Algorithms / Models Used

Algorithm / Technique	Purpose
Otsu Thresholding	Separates foreground and background automatically based on grayscale intensity
HSV Color Thresholding	Detects apple regions based on their colour

Algorithm / Technique	Purpose
Morphological Opening	Removes small noise and unwanted regions
Morphological Closing	Fills small holes and improves the continuity of the apple mask
Bitwise Masking	Extracts the detected apple region
IoU / Jaccard	Measures overlap between predicted and ground-truth masks
Dice Score	Measures similarity between predicted and ground-truth masks
Pixel Accuracy	Measures correctly classified pixels

7. RESULTS

The two segmentation methods were evaluated using the available ground-truth masks.

Metric	Otsu Thresholding	HSV + Morphology
IoU (Jaccard)	23.71%	44.54%
Dice Score	35.79%	51.11%
Pixel Accuracy	30.60%	60.10%

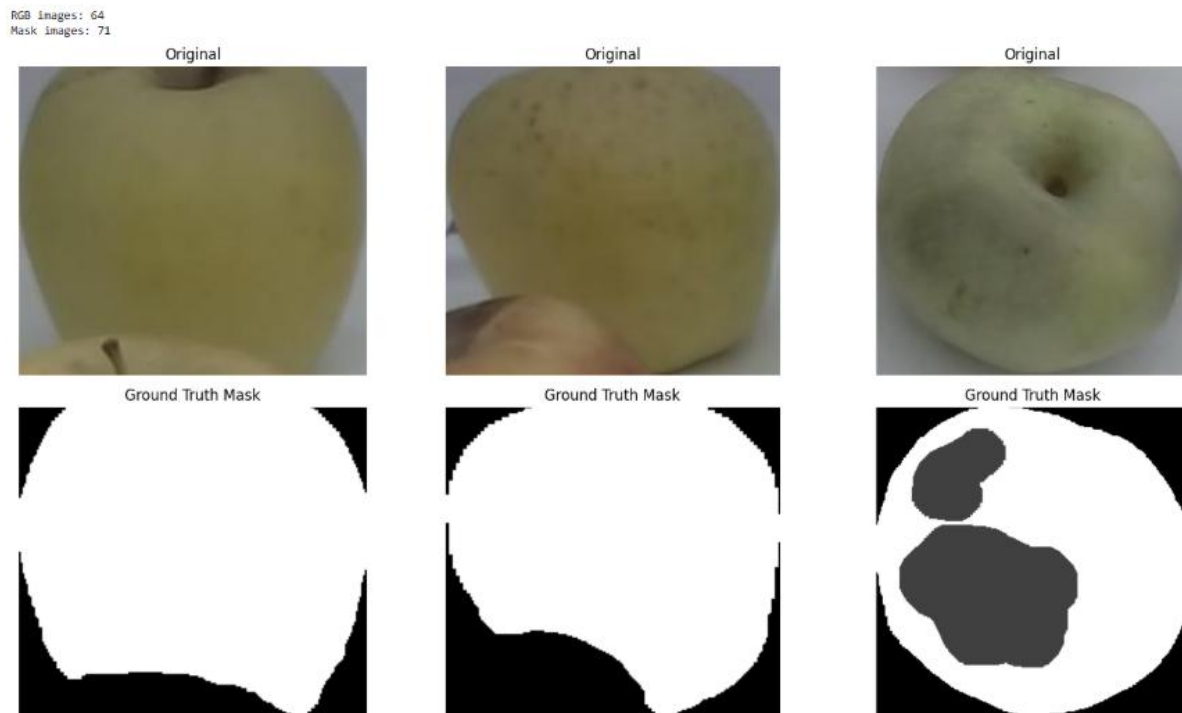
Result Interpretation

The HSV-based method with morphological processing obtained higher scores than Otsu thresholding on the evaluated dataset.

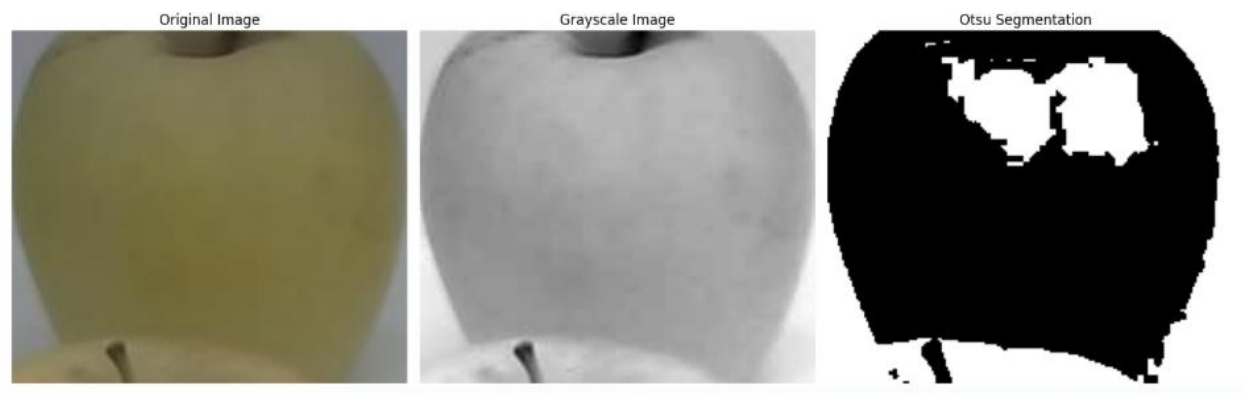
- The IoU increased from 23.71% to 44.54%.
- The Dice Score increased from 35.79% to 51.11%.
- Pixel Accuracy increased from 30.60% to 60.10%.

These results indicate that HSV colour-based segmentation with morphological processing produced better overlap with the ground-truth masks for the tested images.

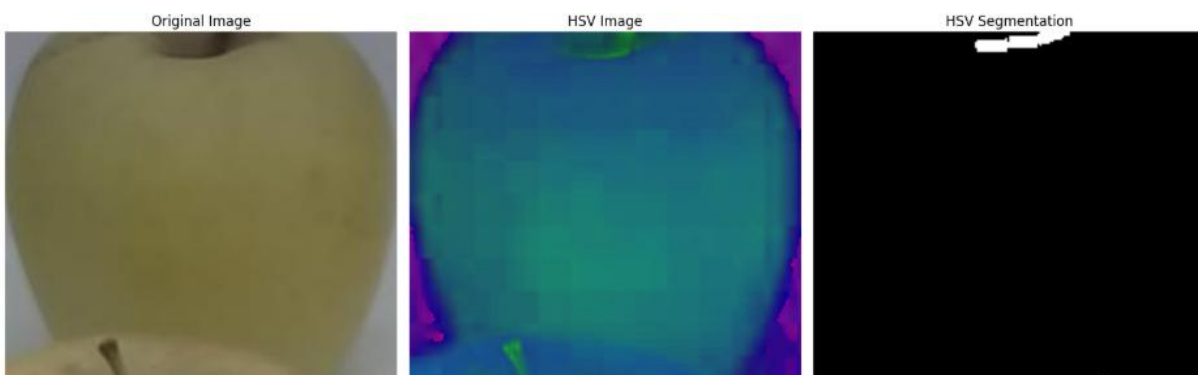
However, the performance depends on image colour, lighting, background, and the selected HSV threshold values.



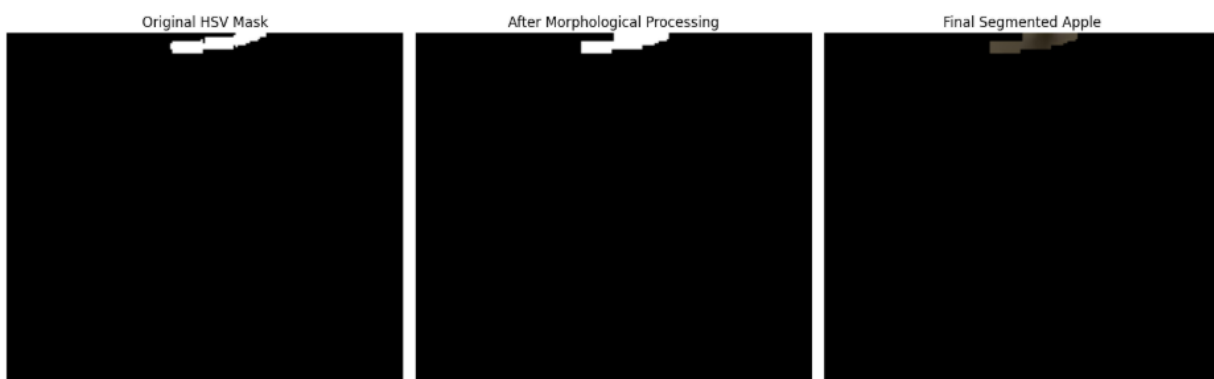
Original and ground truth mask image



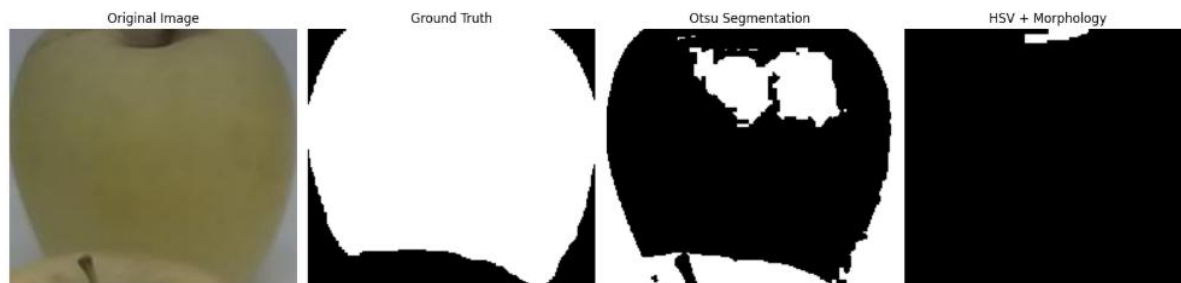
Otsu segmentation



HSV segmentation



Morphologically processed mask



Comparison

Wild JPG images: 7



Image: IMG_0018.jpg

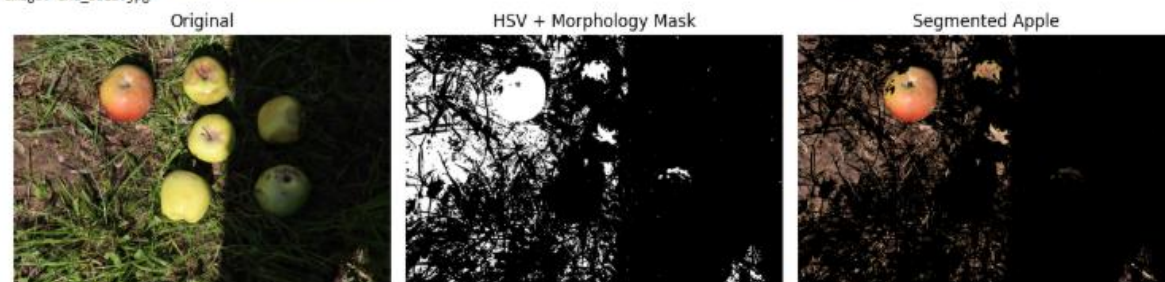
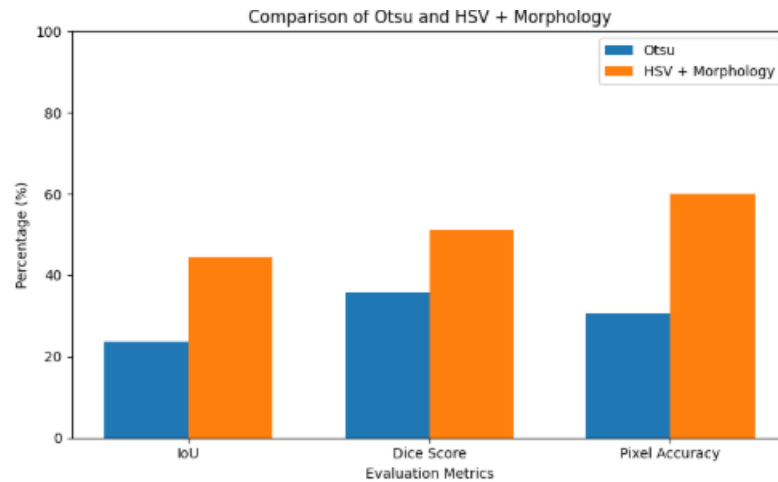


Image: IMG_0126.jpg

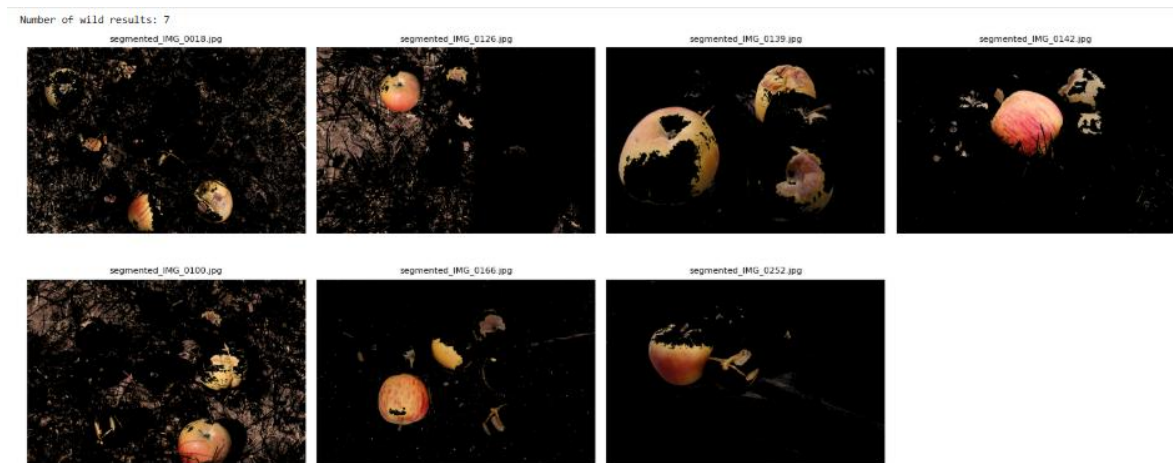


Image: IMG_0139.jpg

Wild images



Comparison graph



Final segmented image

Wild Image Testing

The HSV-based segmentation method was also tested on wild apple images.

Since ground-truth masks were not available for the wild images, IoU, Dice Score, and Pixel Accuracy could not be calculated for this set. The results were evaluated visually.

The wild image testing helped to observe how the method behaves under:

- Natural lighting conditions
- Different backgrounds

- Shadows
- Background colour variations
- Different apple appearances

8. LIMITATIONS

The major limitations of the project are:

- The HSV threshold values were selected manually.
- The method mainly focuses on red and orange apple regions.
- It may not work accurately for green or yellow apples.
- The method is sensitive to lighting conditions.
- Shadows may be incorrectly included in the apple mask.
- Background regions with colours similar to the apple may be detected incorrectly.
- Otsu thresholding may not separate the apple and background effectively when their grayscale intensities are similar.
- Morphological operations may remove small apple details or change object boundaries.
- The wild images do not have ground-truth masks.
- The dataset may not represent all real-world apple images.
- Pixel Accuracy alone is not sufficient to judge segmentation quality because background pixels may dominate the image.
- The method may perform poorly in complex backgrounds or low-contrast images.

9. FUTURE SCOPE

The project can be improved in the following ways:

- Use a larger and more diverse apple image dataset.
- Include red, green, yellow, and partially damaged apples.
- Automatically optimize HSV threshold values.
- Use adaptive thresholding techniques.
- Apply advanced segmentation methods such as watershed segmentation.
- Use edge detection and contour-based processing to improve object boundaries.
- Apply background removal techniques before segmentation.
- Use machine learning or deep learning models such as U-Net for more accurate segmentation.
- Collect ground-truth masks for wild apple images.
- Test the system under different lighting and background conditions.
- Use additional evaluation metrics such as precision, recall, and F1-score.
- Develop a real-time apple segmentation application.
- Deploy the system as a web or mobile application for agricultural and fruit-quality inspection.

10. CONCLUSION

This project developed an apple fruit segmentation system using digital image processing techniques. Two methods, Otsu's thresholding and HSV colour-based thresholding, were implemented and compared.

The HSV-based method was combined with morphological opening and closing to reduce noise, fill small holes, and improve the segmentation mask. The methods were evaluated using IoU, Dice Score, and Pixel Accuracy.

On the evaluated dataset, HSV-based segmentation with morphological processing achieved higher results than Otsu thresholding, obtaining an IoU of 44.54%, a Dice Score of 51.11%, and Pixel Accuracy of 60.10%.

The project helped in understanding colour spaces, thresholding, binary masks, morphological operations, image segmentation, evaluation metrics, and result comparison. It also showed that traditional image-processing methods can be useful for simple segmentation tasks, but more advanced techniques may be required for complex real-world images.